Comparison of linear relationships

1 For each of the following pairs of linear functions, identify which function has the greater y-intercept:

a

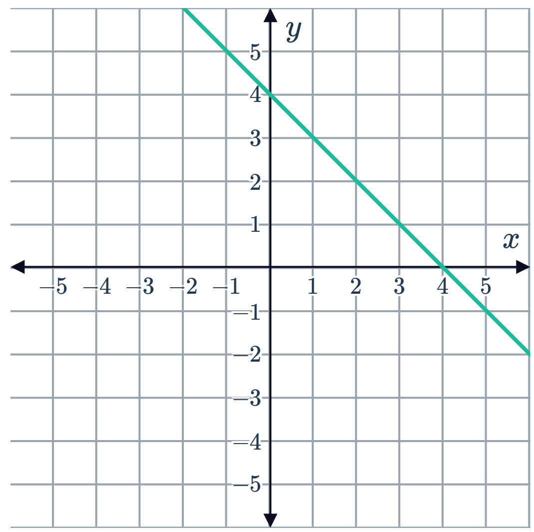
- Function 1: The line with a slope of 4 that crosses the y-axis at $(0, 6)$.
- Function 2: $y = x + 4$

b

- Function 1:

x	2	4	6
y	2	-2	-6

- Function 2:



c

- Function 1:

x	2	4	6
y	19	35	51

- Function 2: $y = 4x + 6$

d

- Function 1:

x	2	4	6
y	19	31	43

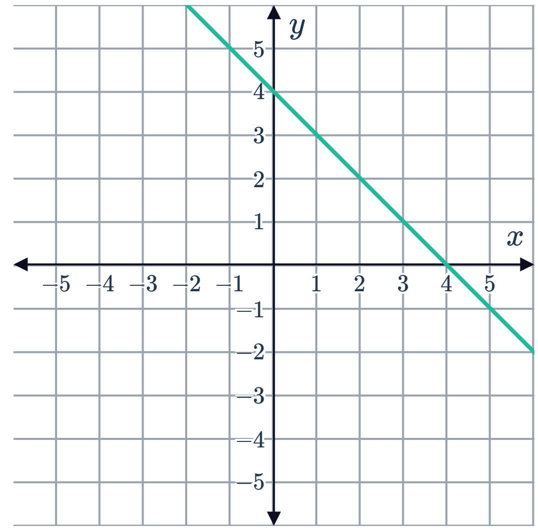
- Function 2: $y = 3x + 4$

e

- Function 1:

x	2	4	6
y	-5	-13	-21

- Function 2:



f

- Function 1: The line with a slope of 7 that crosses the y-axis at $(0, 5)$.
- Function 2: $y = (-m)\,x + 7$

2 For each of the following pairs of linear functions, identify which function has the greater x-intercept:

a

- Function 1: The line with a slope of 3 and a y-intercept of -3.
- Function 2: $y = x - 3$

b

- Function 1: The line with a slope of -1 and a y-intercept of 4.
- Function 2:

x	0	1	2
y	-6	-3	0

c

- Function 1: $y = 5x - 15$
- Function 2:

x	0	1	2
y	6	5	4

d

- Function 1:

x	0	1	2
y	4	2	0

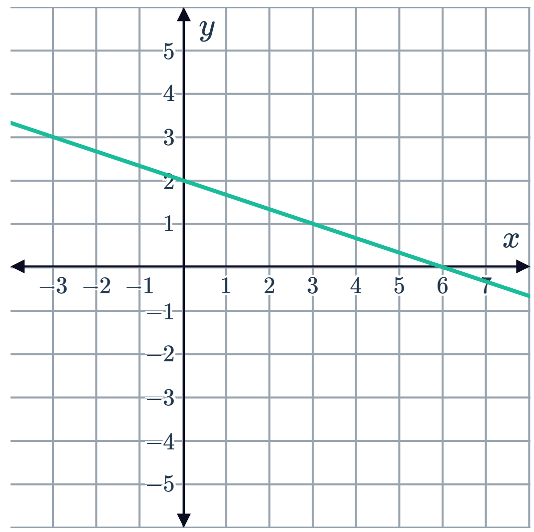
- Function 2: $y = -4x + 16$

e

- Function 1:

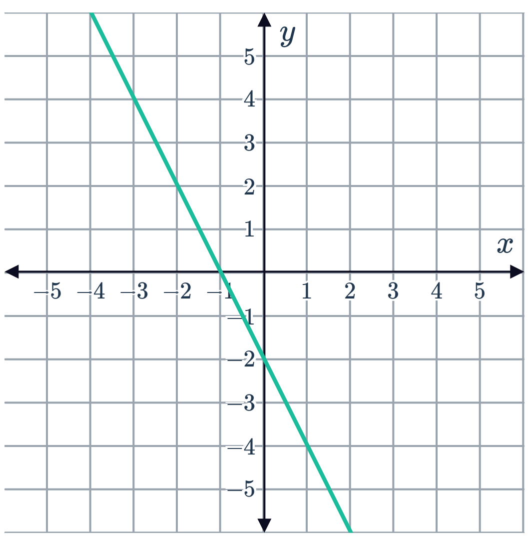
x	0	1	2
y	3	2	1

- Function 2:



f

- Function 1:

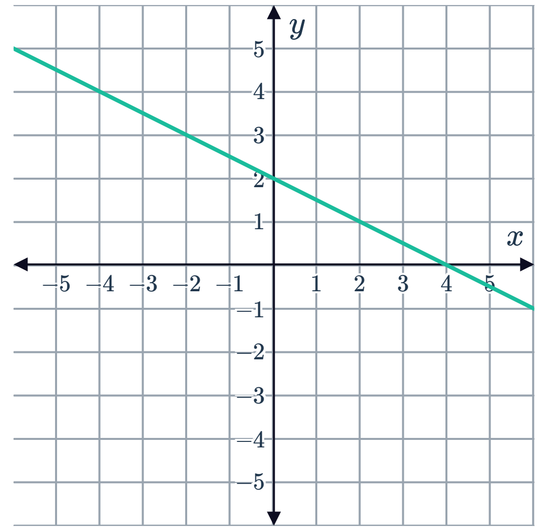


- Function 2:

x	0	1	2
y	−8	−4	0

g

- Function 1: $y = 2x - 2$
- Function 2:



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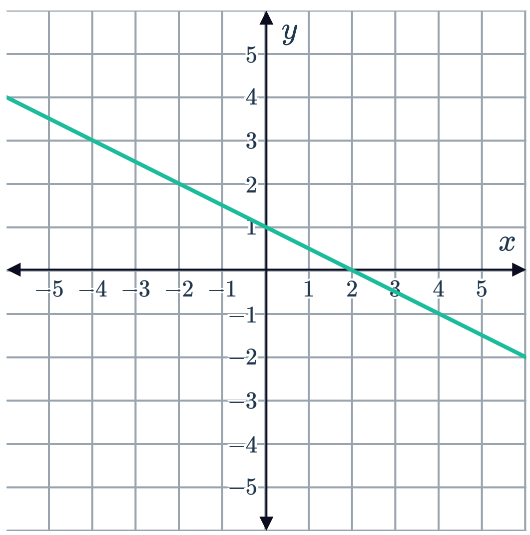
- Function 1:



- Function 2: $y = 2x - 10$

3 For each of the following pairs of linear functions, identify which function has the smaller x-intercept:

a

- Function 1:

x	0	1	2
y	-5	-4	-3

- Function 2: $y = -3x + 6$

b

- Function 1:

x	0	1	2
y	-15	-10	-5

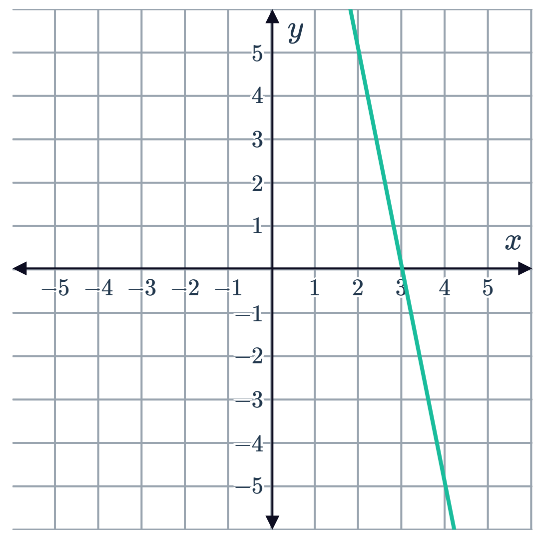
- Function 2: $y = 4x - 24$

c

- Function 1:

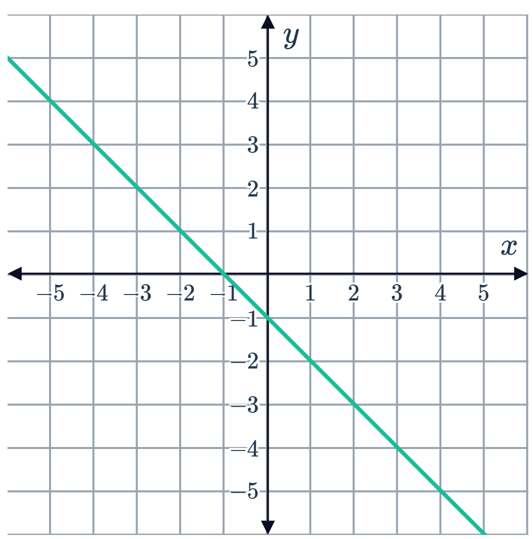
x	0	1	2
y	-25	-20	-15

- Function 2:



d

- Function 1:

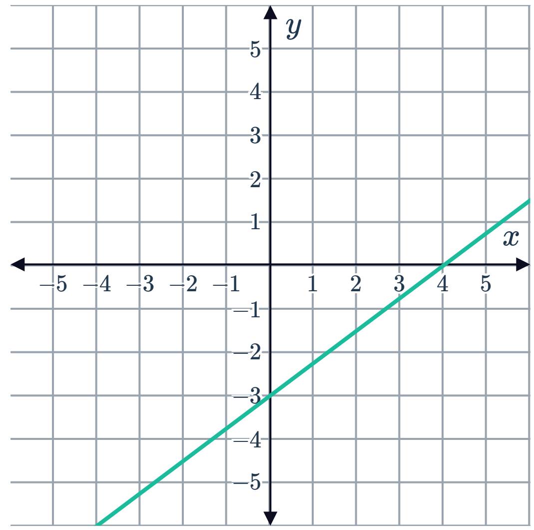


- Function 2:

x	0	1	2
y	12	16	20

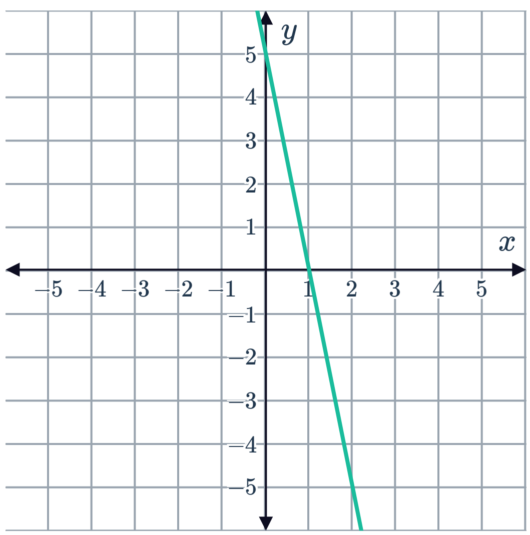
e

- Function 1: $y = -3x + 3$
- Function 2:



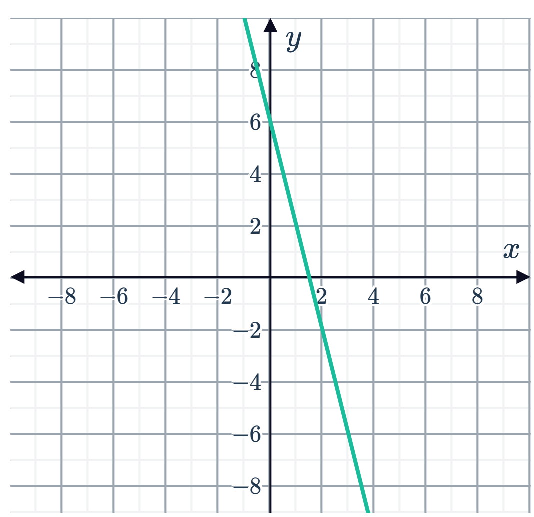
f

- Function 1:



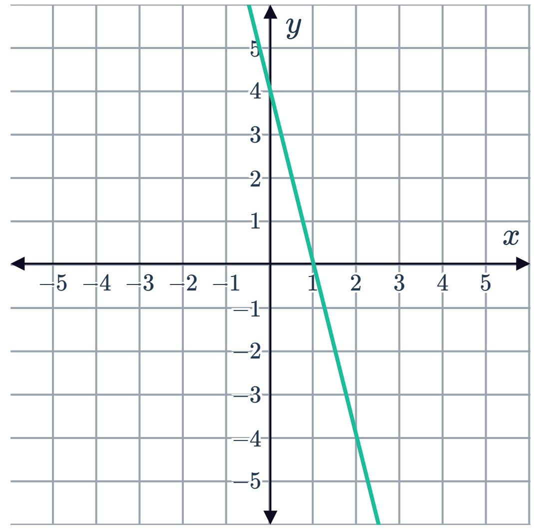
- Function 2: $y = -x + 4$

4 Consider the line shown in the following graph:
Does the line $y = -4x + 4$ have the greater or smaller y-intercept than the graphed line?



5 Consider the line shown in the following graph:

Does the line $y = -4x + 6$ have a greater or smaller y-intercept than the graphed line?



6 Which of the following linear relationships has the smaller corresponding x-value when $y = 42$?

- Function 1:

x	0	1	2
y	2	10	18

- Function 2: $y = 3x + 18$

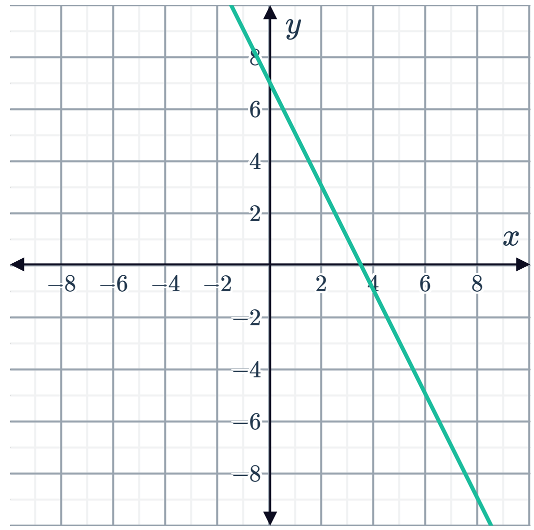
7 Which of the following linear relationships has the smaller corresponding x-value when $y = 46$?

- Function 1: $y = 2x + 32$
- Function 2:

x	0	1	2
y	6	10	14

8 Which of the following linear relationships [redacted] x-intercept?

- Function 1: $y = 3$
- Function 2:



9 For each of the following pairs of linear functions, identify which function has the greater rate of change in the value of y:

a

- Function 1: $y = 1 + 8x$
- Function 2:

x	0	1	2
y	1	6	11

b

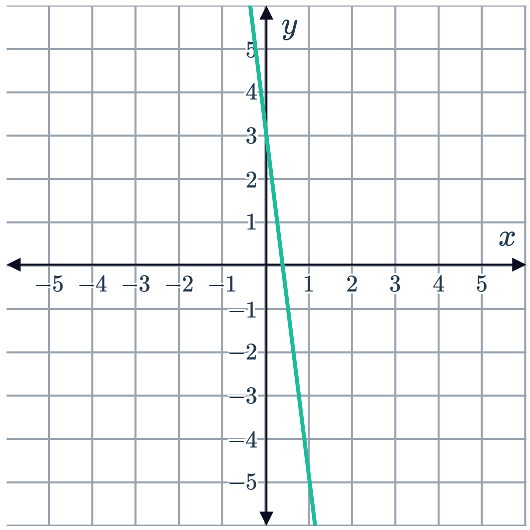
- Function 1: $y = -3 + 3x$
- Function 2:

x	0	1	2
y	-3	2	7

10 Which of the following linear relationships r[redacted]ents a more rapid change in the value of y?

a

- Function 1:



- Function 2:

x	0	1	2
y	3	8	13

b

- Function 1: $y + 5x = 3$

- Function 2:

x	0	1	2
y	3	6	9

11 Which of the following linear relationships represents a slower change in the value of y?

- Function 1: $y + 7x = 3$

- Function 2:

x	0	1	2
y	3	7	11

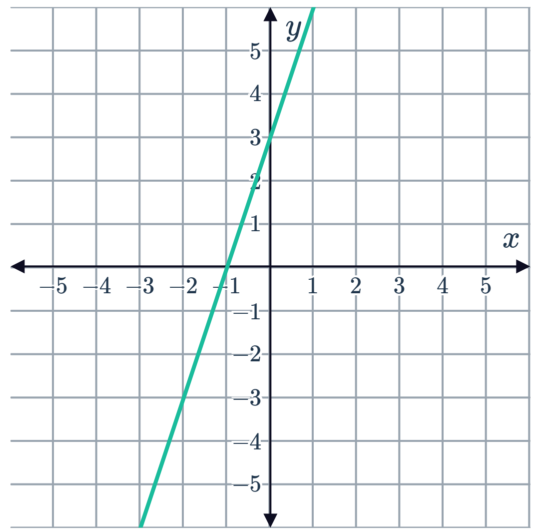
12 For each of the following pairs of linear functions, identify which function has the value of y increasing faster:

a

- Function 1:

x	0	1	2
y	3	7	11

- Function 2:

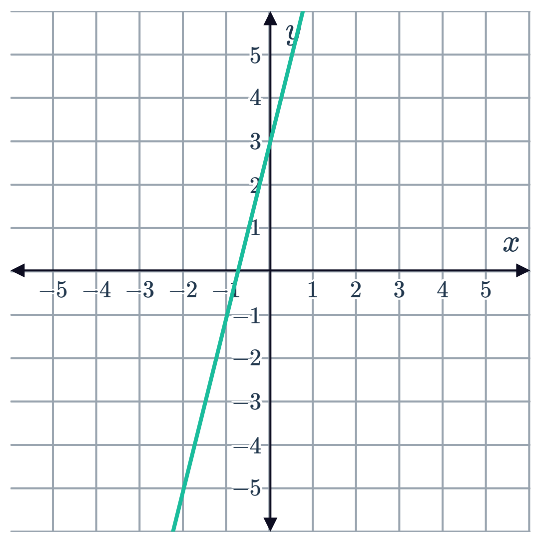


b

- Function 1:

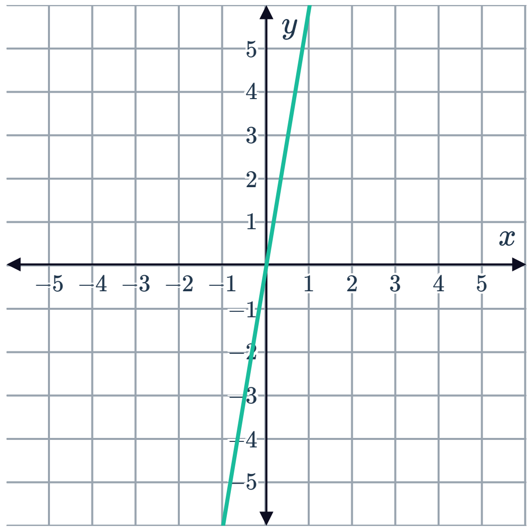
x	0	1	2
y	3	10	17

- Function 2:



c

- Function 1:



- Function 2:

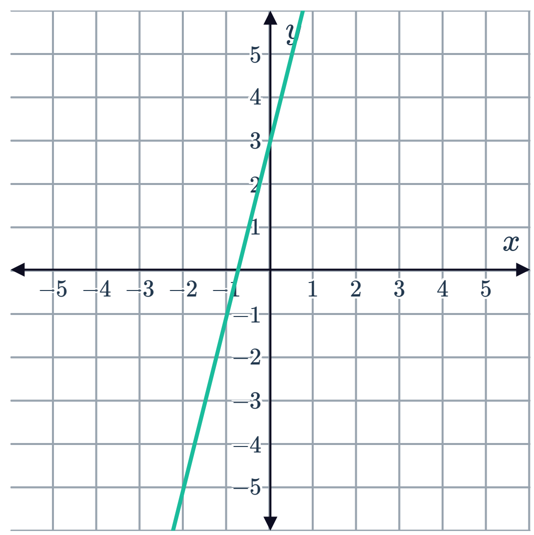
x	0	1	2
y	0	4	8

d

- Function 1:

x	0	1	2
y	-3	4	11

- Function 2



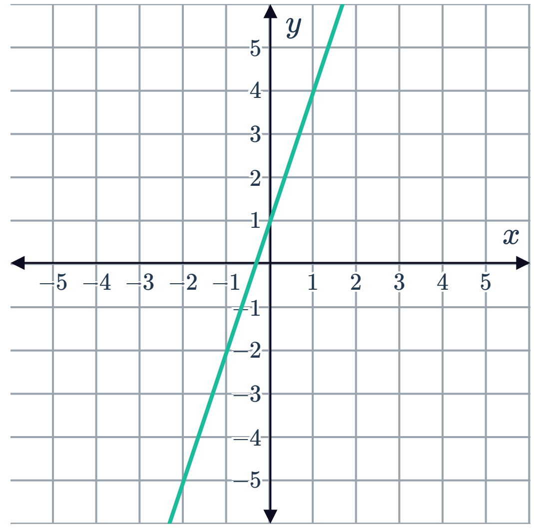
e

- Function 3:

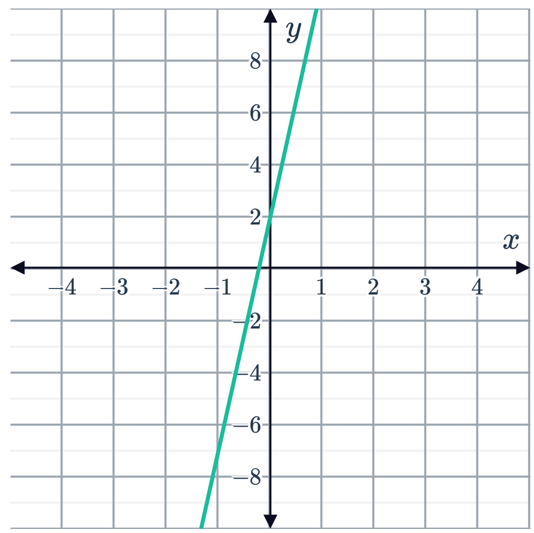
x	0	1	2
y	1	6	11

- Function 4: $y = 8x + 1$

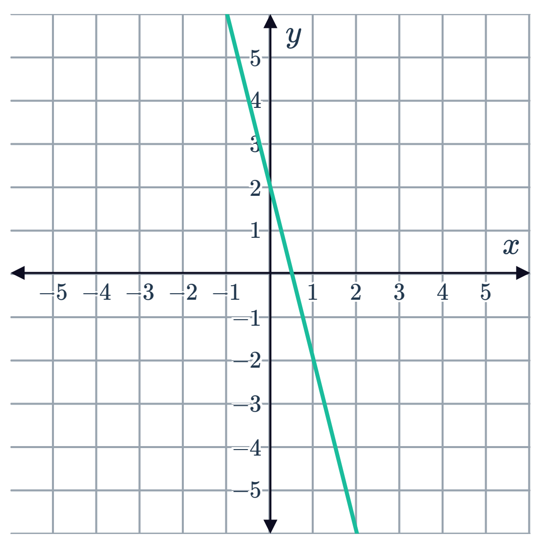
13 Consider the line shown in the following graph:
Does the line $y = 6x + 1$ have a greater or smaller rate of change than the graphed line?



14 Consider the line shown in the following graph:
Does the line $y = 5x + 2$ have a greater or smaller rate of change?



15 Consider the line shown in the following graph:
Are the y-values of the line $y = 6x + 2$ changing more or less rapidly than in the graphed line?



Applications

16 Avril is looking to find the best deal for an internet service to her home, and is comparing two different internet providers:

- Interpute is offering a $\$25$ per month deal where there is no limit to her internet usage.
- Compunet is offering rates according to the amount of internet usage (measured in GB), as shown in the table.

Usage (GB)	Compunet Cost
5	\$17.50
10	\$20.00
20	\$25.00
50	\$40.00

a If u is the amount of internet usage in a month that would make the two offers cost the same, find u.

b Avril estimates that she will be using about 19 GB a month. Which company offers her the better deal?

17 Some friends decide to go camping for the weekend. They cannot all fit in one car so some of them catch a bus to the campground, which is 450 mi from home. Those in the car started driving at 8:00 am and arrived at the campground at 3:30 pm, driving at a constant speed. The bus also drives at a constant speed and takes the same route as the car. Its distance in miles, y, from home x hours after leaving is given by the equation $y = 71x$.

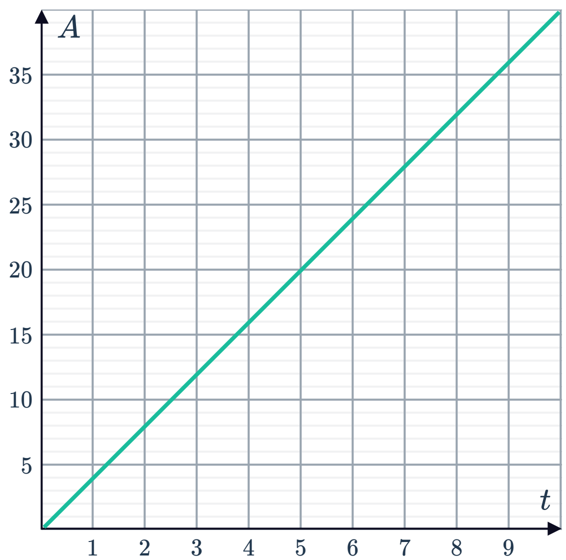
a Determine the speed of the car, in miles per hour.

b Determine the speed of the bus, in miles per hour.

c Determine which vehicle was traveling faster.

18 Mario wants to determine which of two slow[redacted]ase pain medications is more rapidly absorbed by the body.
Consider the graph and table below, which show the amount of medication in the bloodstream for the liquid and capsule form of the medication:

Liquid form



Capsule form

Time (mins), t	Amount in blood (mgs), A
4	24.6
7	42.3
10	60
13	77.7

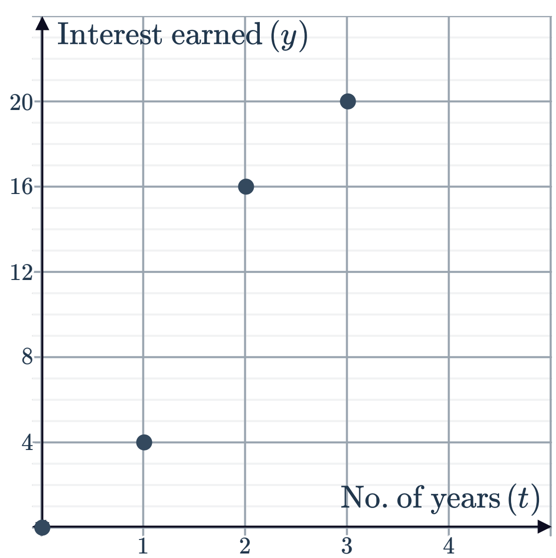
a. Use the graph to find the rate, in mg per minute, in which the liquid form is absorbed.

b. Use the table to find the rate, in mg per minute, in which the capsule form is absorbed.

c. In which form is the medication absorbed more rapidly?

19 Amy and Vanessa have used two different investment strategies to enhance their savings. The amount of interest each has earned after t years is presented in the table and graph:

Amy's interest earned



Vanessa's interest earned

Number of years (t)	0	1	2	3
Interest earned (y)	0	4	8	12

Under whose investment strategy was the interest earned proportional to the number of years passed? Explain your answer.